

Assessing SAM3 for Zero-Shot Building Footprint Delineation: A Case Study in Juba, South Sudan

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Abstract. Accurate and timely building footprint datasets are critical for urban planning and climate vulnerability assessment; however, generating ground truth annotations for model development remains a major bottleneck, especially in data-scarce regions. Recent advancements in vision foundation models have emerged as a promising solution by enabling zero-shot segmentation in label-scarce settings. In this work, we present a systematic evaluation of the Segment Anything Model 3 (SAM3) for building footprint delineation from high-resolution satellite imagery over Juba, South Sudan. We analyze the effects of image resolution, inference window size, and text prompts on building instance segmentation performance. We show that SAM3 achieves an F1 score of up to 0.67 in a zero-shot setting, demonstrating its potential as a scalable solution for mapping buildings with limited labeled data. Our results provide a practical zero-shot baseline for large-scale building delineation, motivating future work for data-efficient model adaptation in the Global South.

Keywords: Computer vision · Remote sensing · Urban mapping

1 Introduction

Accurate and up-to-date building footprint datasets are critical for climate vulnerability assessment and urban planning. These datasets support the analyses of population and infrastructure exposure to climate hazards, inform disaster risk management, and help evaluate the potential impacts of development investments. While initiatives such as Google Open Buildings [12], Microsoft Building Footprints [5], and OpenStreetMap [8] have significantly improved access to large-scale building footprint data, their coverage, completeness, and/or update frequency may not be sufficient for local governments that require timely information on rapidly changing urban environments. This is particularly important in fast-growing cities and conflict-affected regions across the Global South, where settlements can emerge and expand within a short period of time.

To address the growing demand for timely building footprint data, initiatives such as the Smart Parcel Atlas for Resilient Cities (SPARC) by the World Bank

aim to combine high-resolution satellite imagery with machine learning (ML) to map and monitor urban growth in Global South cities. However, generating ground truth data for model training and evaluation remains a major bottleneck in producing ML-generated building footprint datasets. Recent advances in vision foundation models have the potential to alleviate this challenge by enabling building instance segmentation with little to no training data. One such model is the Segment Anything Model 3 (SAM3) [1], a unified vision foundation model for instance segmentation using visual and text prompts. In this work, we investigate the extent to which SAM3 can generate building footprints in a zero-shot setting, with Juba, South Sudan as our primary case study. Specifically, we evaluate the performance of SAM3 across different satellite image resolutions, inference window sizes, and text prompts to better understand its capabilities and limitations for large-scale building footprint mapping.

This study serves as an initial step toward a broader study on data-efficient building footprint generation using vision foundation models. Future work will investigate parameter-efficient fine-tuning of SAM3 using Low-Rank Adaptation (LoRA) [3], as well as data-efficient sampling strategies to minimize the annotation effort required for model adaptation to local contexts in the Global South.

2 Methodology

Dataset. All experiments are conducted on 30 cm/px and 50 cm/px high-resolution satellite imagery from Vantor and Airbus, respectively, covering Juba, South Sudan. In total, 578,634 building footprint polygons were manually digitized by OpenMap Development Tanzania (OMDTZ) [7]. These annotations are used as ground truth for evaluating building instance segmentation models.

Although this study focuses on zero-shot evaluation, we also establish training and validation sets to support future experiments on parameter-efficient fine-tuning and data-efficient model adaptation. To ensure that a diverse range of building characteristics was represented in each data split, the study area was first divided into 600×600 m tiles and partitioned into training, validation, and test sets (80/10/10) using stratified random sampling to preserve the diversity of urban forms across the study area. To characterize urban morphology, each 600 m tile was subdivided into 100×100 m grid cells, for which we computed (i) building coverage ratio (BCR), defined as the total building footprint area divided by the grid area, and (ii) mean building size, defined as the average footprint area of buildings within the cell. Grid cells were then classified into low-, medium-, and high-density categories using BCR thresholds of 0.1 and 0.3, and into small-, medium-, and large-building categories using mean building size thresholds of 50 m^2 and 100 m^2 , yielding nine morphology classes (e.g., low-density, large buildings; low-density, medium-sized buildings; etc.).

For each 600 m tile, the proportions of these classes were used for stratification, ensuring that the distribution of urban forms was approximately preserved across all data splits. The test set, consisting of 55,269 building footprints, is used for all zero-shot SAM3 evaluations in this work, while the training and validation sets are reserved for future model adaptation experiments.

Methods. We begin by exploring zero-shot building delineation using SAM3 [1]. Unlike previous works that focus on semantic segmentation [11, 15], we model the problem primarily as an instance segmentation task, in line with The World Bank’s objectives. Instance segmentation produces individual building footprint polygons, enabling the estimation of building-level attributes and neighborhood statistics such as building counts, density, and size, which are critical for climate resilience assessments, especially in densely built cities across the Global South.

To assess the effectiveness of SAM3 for building delineation, we systematically evaluate the effects of different image resolutions, inference window sizes, and text prompts, including *roof*, *building*, and *house*. Inference is performed via a sliding-window approach with 50% overlap, with inference window side lengths of 45, 75, 150, and 300 m, corresponding to 150, 250, 500, and 1000 px for 30 cm imagery and 90, 150, 300, and 600 px for 50 cm imagery. All experiments were conducted using the SamGeo Python library [9, 14]. Predictions from overlapping windows are merged by grouping polygons with at least 50% overlap using connected components, implemented with NetworkX [2] and dissolving each group into a single polygon. Finally, the resulting building footprints are regularized using the Building-Regularizer Python library to simplify polygon geometries and align edges to their principal direction [13]. All experiments were run using a single NVIDIA GeForce RTX 5090 GPU.

Evaluation. To evaluate model performance under realistic deployment conditions, we apply the full end-to-end SAM3 inference pipeline over all 600×600 m image scenes in the test set using a sliding-window approach. By processing large scenes, we evaluate the robustness of SAM3 to tiling artifacts and associated pre- and post-processing steps, reflecting real-world deployment conditions, similar to the evaluation protocol of [6]. The resulting polygon predictions are then evaluated using deployment-oriented building instance segmentation metrics.

To assess building instance segmentation, we follow the evaluation protocol of the COCO dataset [4], as was done in [10], modified for polygon-level detections. We measure the Intersection over Union (IoU) between two polygons as

$$\text{IoU} = \frac{\text{area}(B_p \cap B_{\text{gt}})}{\text{area}(B_p \cup B_{\text{gt}})} \quad (1)$$

where B_p is the building polygon prediction and B_{gt} is the ground truth polygon. The IoU threshold is used to determine true positives, i.e., for an IoU threshold of 0.5, a detected polygon is classified as a true positive if it overlaps with a ground-truth polygon by at least 50%. From this, we calculate standard evaluation metrics such as the F1-score (F1), Precision (P), Recall (R). As per the COCO benchmark, we also report Average Precision (AP), Average Recall (AR) and average F1 (AF1), calculated over multiple IoU thresholds in the interval from 0.5 to 0.95 with step size 0.05.

3 Results & Discussion

We present the results of our evaluation in Table 1. Overall, zero-shot SAM3 achieves the best F1-score of 0.67 using 30 cm/px satellite image, an inference

window of 75 m (250 px), and the text prompt *roof* for buildings in Juba, South Sudan. Moreover, we note that SAM3 tends to produce high-precision, low-recall outputs; that is, predicted buildings tend to be correct, but many buildings remain undetected. We also observe a substantial performance gap between 30 cm/px imagery (0.67 F1) and 50 cm/px imagery (0.20 F1). However, we note that discrepancies in the geographic co-registration of the 30 cm/px and 50 cm/px imagery may account for the lower performance on the latter.

Across image resolutions and inference window sizes, the text prompt *roof* consistently achieves the highest performance. We also observe that smaller inference windows generally improve segmentation accuracy, with a 75 m window yielding the best results across image resolutions, albeit with relatively longer run times (48 s per tile). However, performance deteriorates for very small inference windows (i.e., 45 m), likely due to insufficient spatial context.

Table 1: SAM3 performance across image resolutions, inference window sizes, and text prompts. F1, P, and R are calculated at an IoU threshold of 0.50. AR, AP and AF1 are computed by averaging values obtained over IoU thresholds 0.50 to 0.95 with a step size of 0.05. Runtime is reported as the mean inference time per 600 m tile.

Size (m)	Size (px)	Prompt	F1 _{0.50}	P _{0.50}	R _{0.50}	AP	AR	AF1	Runtime
Resolution = 0.30 m/px									
45	150	building	0.459	0.578	0.380	0.0937	0.163	0.196	1 min 54 s
		house	0.476	0.563	0.412	0.101	0.174	0.201	
		roof	0.530	0.601	0.475	0.138	0.231	0.258	
75	250	building	0.588	0.690	0.512	0.151	0.232	0.266	48 s
		house	0.601	0.677	0.540	0.161	0.243	0.270	
		roof	0.674	0.744	0.616	0.225	0.307	0.336	
150	500	building	0.559	0.801	0.429	0.135	0.195	0.253	17 s
		house	0.607	0.800	0.489	0.154	0.220	0.273	
		roof	0.625	0.851	0.494	0.180	0.239	0.303	
300	1000	building	0.252	0.809	0.149	0.042	0.064	0.107	6 s
		house	0.323	0.818	0.201	0.055	0.084	0.135	
		roof	0.263	0.874	0.155	0.052	0.071	0.120	
Resolution = 0.50 m/px									
45	90	building	0.103	0.191	0.070	0.004	0.021	0.031	2 min 4 s
		house	0.128	0.189	0.097	0.005	0.029	0.039	
		roof	0.139	0.228	0.100	0.008	0.032	0.044	
75	150	building	0.164	0.254	0.122	0.009	0.039	0.053	50 s
		house	0.182	0.242	0.146	0.010	0.046	0.057	
		roof	0.201	0.277	0.158	0.015	0.052	0.066	
150	300	building	0.161	0.306	0.110	0.011	0.036	0.054	16 s
		house	0.184	0.297	0.134	0.0120	0.043	0.060	
		roof	0.195	0.331	0.138	0.0167	0.047	0.067	
300	600	building	0.090	0.400	0.051	0.007	0.018	0.032	6 s
		house	0.113	0.365	0.067	0.007	0.022	0.038	
		roof	0.120	0.444	0.070	0.011	0.025	0.043	

4 Conclusions

We have presented a systematic evaluation of SAM3 for zero-shot building footprint delineation using high-resolution 30 cm/px and 50 cm/px satellite imagery over Juba, South Sudan. Our results indicate that SAM3 can achieve an F1 score of up to 0.67, with a tendency to produce high-precision, low-recall outputs. Overall, this work establishes a practical zero-shot baseline for building footprint delineation and serves as an initial step for future, broader work on parameter-efficient fine-tuning (e.g., using LoRA [3]) and data-efficient model adaptation within Global South contexts.

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